

Problem 1

a)

We consider the state with basis $|0\rangle, |1\rangle, \dots, |d-1\rangle$, and have another orthonormal basis $|v_0\rangle, |v_1\rangle, \dots, |v_{d-1}\rangle$. To show that there is a unitary transformation U such that $U|i\rangle = |v_i\rangle$ we start with an operator $|v_i\rangle\langle i|$ and check how it acts on a vector $|j\rangle$.

$$(|v_i\rangle\langle i|)|j\rangle = |v_i\rangle\langle i|j\rangle = \langle i|j\rangle|v_i\rangle = \delta_{ij}|v_i\rangle \quad (1)$$

This gives that the operator $|v_i\rangle\langle i|$ will return the vector $|v_i\rangle$ when applied to $|i\rangle$ and 0 otherwise. Taking advantage of this attribute, we construct the transformation by taking

$$U = \sum_{i=0}^{d-1} |v_i\rangle\langle i| \quad (2)$$

To show that U is a unitary operator we check

$$UU^\dagger = \sum_{i=0}^{d-1} |v_i\rangle\langle i| \sum_{j=0}^{d-1} |j\rangle\langle v_j| = \sum_{i,j=0}^{d-1} |v_i\rangle\langle i|j\rangle\langle v_j| = \sum_{i,j=0}^{d-1} |v_i\rangle\delta_{ij}\langle v_j| = \sum_{i=0}^{d-1} |v_i\rangle\langle v_i| = I$$

b)

We see what the operator does to an arbitrary vector of the basis: $\sum_{i=0}^{d-1} c_i |v_i\rangle$

$$\sum_{i=0}^{d-1} |v_i\rangle\langle v_i| \sum_{j=0}^{d-1} c_j |v_j\rangle = \sum_{i=0}^{d-1} \sum_{j=0}^{d-1} c_j |v_i\rangle\langle v_i|v_j\rangle = \sum_{i=0}^{d-1} \sum_{j=0}^{d-1} \delta_{ij} c_j |v_i\rangle = \sum_{i=0}^{d-1} c_i |v_i\rangle\langle v_i| \quad (3)$$

i.e

$$\sum_{i=0}^{d-1} |v_i\rangle\langle v_i| = I \quad (4)$$

Problem 2

a)

Show that the matrix U is unitary, i.e. $U^\dagger U = I$

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix} \quad (5)$$

$$U^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} \quad (6)$$

$$U^\dagger U = UU^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 1*1 + i*(-i) & 1*1 + i*i \\ 1*1 + (-i)*(-i) & 1*1 + i*(-i) \end{pmatrix} = I \quad (7)$$

b)

How U acts upon the states: $|0\rangle, |1\rangle, |+\rangle, |-\rangle, |+i\rangle$ and $|-i\rangle$ we get:

$$U|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle \quad (8)$$

$$U|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ -i \end{pmatrix} = i|-\rangle \sim |-\rangle \quad (9)$$

$|+\rangle$ and $|-\rangle$ is wrong?

$$U|+\rangle = \frac{1}{2} \begin{pmatrix} 1+i \\ 1-i \end{pmatrix} = |+i\rangle \quad (10)$$

$$U|-\rangle = \frac{1}{2} \begin{pmatrix} 1-i \\ 1+i \end{pmatrix} = |1\rangle |-i\rangle \quad (11)$$

$$U|+i\rangle = \frac{1}{2} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle \quad (12)$$

$$U|-i\rangle = \frac{1}{2} \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle \quad (13)$$

We see that the unitary operator U transforms $|0\rangle \rightarrow |+\rangle, |+\rangle \rightarrow |i\rangle, |i\rangle \rightarrow |0\rangle$, i.e. $z \rightarrow x, x \rightarrow y, y \rightarrow z$

c)

U acts on the bloch sphere with a 120° rotation around the $(1, 1, 1)$.

Problem 3

Let $|0\rangle, |1\rangle, |2\rangle$ be a basis for the system.

a)

Firstly we need to check the unit length of the basis. Using the characteristic $\langle x_i | x_j \rangle = \delta_{ij}, \forall x_{i,j} \in |0\rangle, |1\rangle, |2\rangle$ we get the following inner products:

$$\langle a | a \rangle = \frac{1}{2} \langle 0 | 0 \rangle + \frac{1}{4} \langle 1 | 1 \rangle + \frac{1}{4} \langle 2 | 2 \rangle = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1 \quad (14)$$

$$\langle b | b \rangle = \frac{1}{2} \langle 0 | 0 \rangle + \frac{1}{4} \langle 1 | 1 \rangle + \frac{1}{4} \langle 2 | 2 \rangle = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1 \quad (15)$$

$$\langle c | c \rangle = \frac{1}{2} \langle 1 | 1 \rangle + \frac{1}{2} \langle 2 | 2 \rangle = \frac{1}{2} + \frac{1}{2} = 1 \quad (16)$$

There can only exist three(3) orthonormal vectors in a three dimensional space, so we test inner products.

$$\langle a | b \rangle = \frac{1}{2} \langle 0 | 0 \rangle - \frac{1}{4} \langle 1 | 1 \rangle - \frac{1}{4} \langle 2 | 2 \rangle = \frac{1}{2} - \frac{1}{4} - \frac{1}{4} = 0 \quad (17)$$

$$\langle b|c\rangle = -\frac{\sqrt{2}-1}{2}\langle 1|1\rangle + \frac{\sqrt{2}-1}{2}\langle 2|2\rangle = -\frac{\sqrt{2}-1}{2} + \frac{\sqrt{2}-1}{2} = 0 \quad (18)$$

$$\langle a|c\rangle = \frac{\sqrt{2}-1}{2}\langle 1|1\rangle - \frac{\sqrt{2}-1}{2}\langle 2|2\rangle = \frac{\sqrt{2}-1}{2} - \frac{\sqrt{2}-1}{2} = 0 \quad (19)$$

i.e. $|a\rangle, |b\rangle, |c\rangle$ is an orthonormal basis.

b)

The von neumann method of measurement is using a self-adjoint operator named an observable to conduct the measurement. We take the projection of the state onto the basis:

$$\langle \psi|a\rangle = \frac{1}{\sqrt{6}}\langle 0|0\rangle + \frac{1}{\sqrt{6}}\langle 1|1\rangle \quad (20)$$

$$\langle \psi|b\rangle = \frac{1}{\sqrt{6}}\langle 0|0\rangle - \frac{1}{\sqrt{6}}\langle 1|1\rangle \quad (21)$$

$$\langle \psi|c\rangle = \frac{2}{\sqrt{6}}\langle 1|1\rangle \quad (22)$$

I.e. we get probability $\frac{1}{3}$ of being in $|a\rangle$, $\frac{1}{3}$ for $|b\rangle$ and $\frac{1}{3}$ of being in state $|c\rangle$.

For measurement of $|a\rangle, |b\rangle$ and $|c\rangle$ the state collapses to subsequently $|a\rangle, |b\rangle$ and $|c\rangle$

What are the resulting states?

Problem 4

a)

Probability of transmission through a polarizer is given by Malus' law: $P = \cos^2(\Delta\theta)$ If we assume completely unpolarized light before the first filter, photons must rotate by 45 degrees three(3) times. the amount of light passing through will be $\cos^2(45^\circ) \cdot \cos^2(45^\circ) \cdot \cos^2(45^\circ) = \frac{1}{8}$

b)

For the next setup, we have two options: [1] : $0^\circ \rightarrow 30^\circ \rightarrow 45^\circ \rightarrow 60^\circ \rightarrow 90^\circ$ or [2] : $0^\circ \rightarrow 60^\circ \rightarrow 45^\circ \rightarrow 30^\circ \rightarrow 90^\circ$.

First we check option [1]:

$$P = \frac{1}{2} \cdot \cos^2(30^\circ) \cdot \cos^2(15^\circ) \cdot \cos^2(15^\circ) \cdot \cos^2(30^\circ) \approx 0.245$$

and option [2]:

$$P = \frac{1}{2} \cdot \cos^2(60^\circ) \cdot \cos^2(15^\circ) \cdot \cos^2(15^\circ) \cdot \cos^2(60^\circ) \approx 0.027$$

So option [1] will let substantially more light through.

Problem 5

a)

if the filter gives 25% regardless of orientation, we know that $\cos^2(\Delta\theta) \cdot \cos^2\left(\frac{\pi}{2} - \Delta\theta\right)$. The first component of each object is a perfectly polarizing filter just the same as earlier. The other side must be a substance that can "un-polarize the light", i.e. take in polarized light and send out the same intensity as unpolarized light. Since the configuration AB CD gives the regular behavior of polarizing filters, sides B and C must be those. The flipping of object CD to DC changes

b)

If I put side A next to side D I get the same as when I had side B next to side D, as $A = D$ and these both project to the same eigenstate. Unpolarizing light two times in a row does not change the light, after the first polarization it is already in an unpolarized eigenstate, and a second operation will just return the same state.